



DR. B. C. ROY ENGINEERING COLLEGE, DURGAPUR

AN AUTONOMOUS INSTITUTE

AFFILIATED TO: MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST
BENGAL

Applied Exponential Models for Electrical Engineering- I & II

EE-AOC-581 & EE-AOC-681

B.Tech (EE) (5th and 6th Semester)

20 Selected Students | Batch B2024-28

*One Curve, Many Worlds: From Circuit Theory to Motor Drive
Add-On Course Syllabus (Two Semester Programme)*

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Academic Year 2026-27

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1 Course Overview

This add-on course addresses a fundamental gap in the standard undergraduate curriculum: the exponential family e^{st} , $s \in \mathbb{C}$, is the single mathematical language that governs every major phenomenon in electrical engineering, yet students encounter each phenomenon in a separate subject silo and rarely perceive the common thread.

The course connects content students have already studied – Basic Electrical, Network Theory, Machines, Control Systems – through this unified mathematical lens, then extends the understanding progressively through power electronics design to motor drive implementation. The course is **not about teaching new content in isolation**. It is about reconnecting and deepening what is already there, one exponential curve at a time.

Course Fundamentals

 **Course Code:** BCREC-EE-AOC-01 (MAKAUT assignment pending)

 **Official Title:** Applied Exponential Models for Electrical Engineering -I & II

 **Course Tagline:** *One Curve, Many Worlds: From Circuit Theory to Motor Drive*

 **Type:** Add-On / Value-Added Course (Non-Credit, Certificated)

 **Total Contact Hours:** 74 Hours (42 in Sem-5 + 32 in Sem-6)

 **Batch Strength:** 20 (Selected students – Batch B2024–28)

 **Delivery Mode:** Integrated Lecture cum Hands-On Lab

 **Lab / Venue:** Sim Lab-III, EE Dept., BCREC (Tesla Research Lab as extended venue if required)

 **Target Batch:** B.Tech EE, 5th Semester (Part-I) and 6th Semester (Part-II), AY 2026–27

 **Compliance:** NBA (11 POs, 3 PSOs), NAAC 2.3/3.4, AICTE Model Curriculum

2 Programme Educational Objectives (PEOs)

 **Note:** PEOs are long-range goals observed 3–5 years after graduation, as per NBA for B.Tech programmes. Three PEOs are defined for this add-on course in alignment with the parent programme PEOs of the B.Tech (EE) programme at BCREC.

PEO 1 – Technical Competence:

Graduates will apply a unified understanding of exponential mathematics and electrical engineering principles to analyse and design power electronic systems, motor drives, and embedded control solutions in professional practice.

PEO 2 – Industry Readiness and Adaptability:

Graduates will demonstrate the ability to read and interpret manufacturer documentation, implement industry-standard control strategies (Type-II/III compensators, FOC), and use professional simulation tools (OpenModelica, LTspice, TI WEBENCH) in employment or entrepreneurship.

PEO 3 – Lifelong Learning and Research Orientation:

Graduates will engage in continuous self-improvement through higher studies, professional certifications, and contributions to the advancement of electrical engineering – particularly in power electronics, motor drives, and smart energy systems.

3 Course Outcomes (COs)

i Anderson and Krathwohl Revised Bloom's Taxonomy is used throughout.

Abbreviations: **Re** = Remember, **Un** = Understand, **Ap** = Apply, **An** = Analyse, **Ev** = Evaluate, **Cr** = Create

Upon successful completion of both semesters, students will be able to:

- CO1 Identify** and **classify** the exponential function family (e^t , e^{-t} , $1-e^{-t}$, sigmoid, tanh, cosh, $e^{j\omega t}$, $e^{(\sigma+j\omega)t}$) and **correlate** each member to its corresponding physical phenomenon in electrical circuits, power systems, and machines. [BL1 Re, BL2 Un]
- CO2 Analyse** the steady-state and transient behaviour of passive networks (R, L, C) and DC–DC converter circuits (Buck and Boost) using Volt-second and Current-second balance principles, state-space modelling, transfer function derivation, and Bode plot interpretation – including ESR zero and non-minimum phase response. [BL4 An]
- CO3 Apply** datasheet reading skills and circuit design principles to **implement** linear voltage regulators (fixed and variable), protection circuits, and gate drive circuits (analog, Arduino/ESP32-based, optically isolated, and UCC27524 IC-based). [BL3 Ap, BL6 Cr]
- CO4 Design** Type-II and Type-III compensators for closed-loop control of Buck and Boost converters using the systematic K-factor Bode-plot method, and **evaluate** the achieved stability margins through simulation and hardware verification. [BL5 Ev, BL6 Cr]
- CO5 Explain** the physical necessity of the abc-to-dq0 (Park) transformation for AC machine control and **apply** field-oriented control (FOC) principles to DC, BLDC, and PMSM drives, identifying the role of each sub-block in the complete drive architecture. [BL2 Un, BL3 Ap]
- CO6 Build** and **evaluate** a complete motor drive system – from gate pulse generation to closed-loop speed control – using OpenModelica simulation and/or Hardware-in-the-Loop (HIL) / Processor-in-the-Loop (PIL) implementation. [BL5 Ev, BL6 Cr]

Table 1: Anderson and Krathwohl Revised Bloom's Taxonomy Coverage

BL Level	Cognitive Process	COs	How Addressed in This Course
BL1	Remember (Re)	CO1	Recall exponential curve types and EE phenomena
BL2	Understand (Un)	CO1, CO5	Explain correlation of math to physics; interpret dq0
BL3	Apply (Ap)	CO3, CO5	Circuit fabrication; FOC block diagram application
BL4	Analyse (An)	CO2	State-space derivation; Bode plot analysis; ESR effects
BL5	Evaluate (Ev)	CO4, CO6	Verify compensator margins; validate HIL results
BL6	Create (Cr)	CO3, CO4, CO6	Design circuits; design compensators; build drive system

4 Graduate Attributes

Note: Graduate Attributes (GAs) are the qualities, skills, and understandings a student should develop during their programme. The following attributes are explicitly cultivated through this course and map directly to NBA Graduate Attributes for B.Tech programmes.

Table 2: Graduate Attributes Developed Through This Course

Sl.	Graduate Attribute	How Developed in This Course	Modules
1	Mathematical Thinking	Exponential family unification; Euler's formula; state-space derivation; Park transformation matrix; K-factor compensator design – all require structured mathematical reasoning rather than formula substitution.	M1, M2, M4, M6, M7
2	Computational Modelling	Python interactive dashboards; Scilab/Xcos block diagrams; LTspice circuit simulation; OpenModelica system-level modelling; HIL/PIL implementation – students build, run, and interpret computational models at every stage.	M2, M4, M6, M7, M8

Continued on next page...

(continued) Graduate Attributes Developed Through This Course

Sl.	Graduate Attribute	How Developed in This Course	Modules
3	Engineering Judgement	Choosing Type-II vs. Type-III compensator; selecting crossover frequency; deciding between analog and digital implementation; interpreting non-minimum phase step response – judgement-based decisions with real hardware consequences.	M4, M5, M6, M7
4	Problem Solving	Each module is structured as a problem-solution narrative: Module 4 identifies the control problem; Module 6 solves it. Module 2 identifies the compartmentalisation problem; the entire course solves it. Students learn to frame and resolve engineering problems systematically.	All Modules
5	Analytical Reasoning	Bode plot interpretation; pole-zero physical interpretation; dq0 transformation derivation and validation; CO attainment viva – every assessment requires reasoning from first principles, not recall of results.	M1, M2, M4, M6, M7

5 Prerequisites

5.1 Formal Subjects (Completed by End of Semester 4)

Table 3: Formal Subject Prerequisites

Sl.	Subject	Relevant Concepts Used in This Course
1	Basic Electrical Engineering	RC/RL transient response, KVL/KCL, Thevenin/Norton, phasors
2	Network Theory	Mesh/node analysis, transient/steady-state, Laplace basics
3	Electromagnetic Fields	Wave propagation $e^{j\omega t}$, field equations, energy in fields
4	Electrical Machines – I	B–H curve (tanh shape), transformer equivalent, DC motor
5	Analog Electronics	Diode I–V (sigmoid shape), op-amp, feedback circuits
6	Measurement and Instrumentation	Lissajous figures on CRO, phase and frequency measurement

5.2 Co-requisite Subjects (Running in Semester 5 – Part I)

Table 4: Co-requisite Subjects Running Concurrently in Semester 5

Sl.	Subject	Connection Exploited in This Course
1	Electrical Machines – II	Induction motor torque-slip curve (sigmoid), PMSM basics
2	Power System – I	Transmission line sag (catenary), P–V nose curve (sigmoid)
3	Control Systems	Root locus, Bode plot, pole-zero response, e^{st} in TF
4	Power Electronics	Converter topologies, switching waveforms, duty cycle
5	EV and Hybrid Vehicles	Battery SoC curve (saturating rise), BESS integration

5.3 Skill Prerequisites

- ✓ Basic familiarity with Python or Scilab for plotting (not programming-level expertise required)
- ✓ Ability to use a laboratory breadboard and digital multimeter
- ✓ Basic oscilloscope operation (covered in Measurement and Instrumentation)
- ✓ Willingness to engage with mathematical notation at engineering level

6 Student Selection Procedure and Course Rules

6.1 Student Selection Procedure

i Philosophy: This course is designed for motivated, self-selected students who demonstrate genuine interest and adequate baseline competence. Selection is competitive and conducted in three transparent stages before the commencement of Semester 5. Maximum final batch strength is **20 students** from Batch B2024–28.

Table 5: Three-Stage Competitive Student Selection Process

Stage	Mode	Tool / Platform	Target	Description
1	Expression of Interest	Google Form (online)	Up to 40 students	Students fill a structured Google Form stating their motivation, available weekly time commitment, and prior exposure to Python, C, or circuit work. No elimination at this stage – shortlisting only.
2	Aptitude Screening Quiz	Google Quiz (online)	Up to 30 students	Online timed quiz covering: (a) basic Python / C programming concepts, (b) fundamental electrical concepts (Ohm's law, KVL, KCL, RC transient, basic logic). Duration: 30 minutes. Minimum qualifying score: 60%. Result communicated within 3 working days.
3	Technical Viva Voce	In-person (EE Dept.)	Final 20 students	Short face-to-face interaction (10 minutes per student) to assess curiosity, communication, and baseline physical understanding. Panel: Course Instructor and HoD EE. Final selection list published on the departmental notice board.

⚠ Selection Timeline: Stage 1 (Google Form) opens **4 weeks** before the start of Semester 5. Stage 2 (Google Quiz) is conducted **2 weeks** before the start of Semester 5. Stage 3 (Viva Voce) is conducted **1 week** before the start of Semester 5. The final batch list is frozen on the first day of Semester 5. No additions are permitted after the first session under any circumstances.

6.2 📋 Course Rules and Regulations

📅 Attendance Policy

Minimum attendance required: 80% of all scheduled contact hours.

Any session falling on a **college-declared holiday or government gazette holiday** will not be counted as a contact hour and will not affect any student's attendance calculation. However, if a student is absent on any other scheduled session day – including days when a group of students collectively chooses not to attend – that session will be marked as **absent without exception**. Collective absence of a group of students on a non-holiday session day carries no special status and will be treated identically to individual absence. There is **no provision** for compensatory sessions, make-up classes, or attendance relaxation on grounds of group absence.

Students falling below 80% attendance in any module will not be permitted to submit that module's lab notebook for assessment. The corresponding CO attainment for that module will be marked as *Not Attained* in the course record.

Attendance is recorded at the start of each session. Late entry (beyond 10 minutes) is not permitted and the student will be marked absent for that session.

📋 General Course Conduct Rules

- ✓ This is a **non-credit, certificated** add-on course. It does not appear in the MAKAUT mark sheet but a departmental certificate of completion is issued upon passing.
- ✓ Lab notebooks must be **handwritten and sketched** (circuit diagrams, waveforms) with Python/Scilab printouts attached. Fully printed submissions are not accepted.
- ✓ All lab sessions require **prior reading** of the relevant module topic. Students who arrive without background preparation may be asked to observe rather than participate in that session.
- ✓ Mobile phones must be **switched off or kept in silent mode** during all sessions.
- ✓ Equipment in Sim Lab-III must be handled with care. Any damage due to negligence is the student's responsibility.
- ✓ **Plagiarism** in lab notebooks or reports will result in zero marks for that module and a note in the student's departmental record.
- ✓ Students who drop out after Stage 3 selection without informing the instructor at least one week in advance forfeit the right to rejoin in a future batch.

⚠ Continuation from Semester 5 to Semester 6: Only students who complete Semester 5 (Part-I) with a minimum aggregate of 50% and attendance of **at least 80%** are eligible to continue into Semester 6 (Part-II). There is no provision for partial re-admission or re-examination. The certificate of completion is issued only on successful completion of **both semesters**.

7 📖 Semester 5 – Part I (42 Contact Hours)

This semester builds the mathematical foundation and power electronics toolkit. The exponential family (Module 2) is the spine; every subsequent module references it explicitly.

7.1 ⚡ Module 1: Network Theorems and Energy Properties of R, L, C (8 Hours)

🚩 Module Objective: Reconnect students with network theory at a deeper physical level and establish Volt-second / Current-second balance as the engineering language used throughout the course. Introduce the complex plane as the natural home of EE mathematics.

🔑 Key Topics:

- Network theorems – Thevenin/Norton in real-world contexts (battery equivalent, multi-source grid)
- R, L, C impedance vs. frequency; phase behaviour; reactance curves
- Volt-second balance of L: $\int V_L dt = 0$ (steady state); physical meaning
- Current-second balance of C: $\int I_C dt = 0$ (steady state); physical meaning
- Euler's formula $e^{j\theta} = \cos \theta + j \sin \theta$; complex plane; phasor as frozen $e^{j\omega t}$
- Impedance locus on complex plane; introduction to pole and zero as complex-plane objects

🔧 Lab / Hands-On: Thevenin equivalent verification with real circuit and oscilloscope; Python/Scilab plots of impedance vs. frequency; phasor diagram and impedance locus Python visualisation

📁 Module 1 Assessment: Lab notebook: circuit measurements and Python plots with physical interpretation paragraph for each result

COs: CO1, CO2 **Bloom Level:** BL1 (Re), BL2 (Un), BL4 (An)

7.2 ⚡ Module 2: The Exponential Family – One Curve, Many Worlds (8 Hours)

🚩 Module Objective: Establish the complete exponential family as the *mathematical spine* of electrical engineering. Every curve is shown to be already present in the student's engineering background – under a different name in a different subject.

🔑 Key Topics:

- Pure growth e^t and pure decay e^{-t} : natural response of RL/RC circuits
- Saturating rise $(1 - e^{-t})$ and saturating drop: capacitor charging; battery SoC (EV course link)
- Sigmoid $\frac{1}{1+e^{-x}}$: induction motor torque-slip curve (Machines-II); diode I-V (Analog); P-V collapse (Power System-I)
- Hyperbolic tangent $\tanh(x)$: B-H magnetisation curve (Machines-I); magnetic core saturation – ML borrowed this from physics
- Catenary $\cosh(x)$: transmission line sag equation (Power System-I); physical derivation sketch
- Damped oscillation $e^{(\sigma+j\omega)t}$: underdamped LC response; Lissajous figure as projection of rotating phasor
- Unification: all curves on one plot; pole location vs. curve shape; time constant as universal parameter

✂ **Lab / Hands-On:** Python interactive dashboard: all curves with slider controls; overlay sigmoid, torque-slip, and diode curves; B-H data fit to $\tanh$; Lissajous figure generation; transmission line sag computation

📁 **Module 2 Assessment:** Lab notebook: one physical interpretation paragraph per curve; identification exercise – given a waveform, identify which exponential family member it belongs to and which subject it came from

COs: CO1, CO2 **Bloom Level:** BL1 (Re), BL2 (Un)

⚠ **Key Pedagogical Point:** The induction motor torque-slip curve, the battery SoC charging curve, and the diode I-V characteristic are all members of the same exponential family. Students who genuinely see this once never compartmentalise EE subjects again. This module is the intellectual heart of the course.

7.3 ⚡ Module 3: Linear Voltage Regulator – Design, Protection, and Fabrication (6 Hours)

🚩 **Module Objective:** Provide the first complete real-world build experience. Establish datasheet reading as a core engineering skill. Bridge from mathematical theory to fabricated, tested, working hardware.

🔑 Key Topics:

- Why a linear regulator? Dropout voltage, efficiency, noise – where it wins over SMPS
- Reading a datasheet live: 7805 – absolute maximum ratings, thermal resistance, ripple rejection
- Fixed regulator circuits: 7805/7812/7905; decoupling capacitor role (Module 1 link)
- Variable regulator LM317/LM337: output voltage equation; R1/R2 calculation

- Protection circuits: current limiting (sense resistor); thermal shutdown (sigmoid characteristic – Module 2 link); overvoltage crowbar; foldback limiting
- Complete bench power supply: integration of all above; layout for minimum noise

✂ **Lab / Hands-On:** Build fixed 5 V supply; measure ripple vs. datasheet; build LM317 variable supply; characterise output vs. R2; implement current limiting; final integration into complete bench supply with LED indicators

📁 **Module 3 Assessment:** Lab notebook: annotated datasheets, measurement tables, circuit photographs, one-paragraph physical explanation of each protection mechanism

COs: CO3 Bloom Level: BL3 (Ap), BL6 (Cr)

7.4 ⚡ Module 4: Switch Mode Power Supply – Buck and Boost Open Loop (10 Hours)

🚩 **Module Objective:** Develop complete analytical understanding of Buck and Boost converters from physical waveforms through state-space to Bode plot. Explicitly treat ESR zero and non-minimum phase response of the Boost converter – the problems that Module 6 will solve.

🔑 Key Topics:

- Buck converter: inductor current waveform; $V_o/V_{in} = D$ from Volt-second balance; CCM vs. DCM
- Buck state-space averaging: state variables i_L , v_C ; small-signal model; $G_{vd}(s)$; Bode plot
- Boost converter: $V_o/V_{in} = 1/(1 - D)$ from Volt-second balance; state-space; $G_{vd}(s)$
- Right-half-plane (RHP) zero in Boost: physical origin – energy storage delay; step response consequence
- ESR of output capacitor: physical meaning; LHP zero it creates; impact on Bode phase curve
- Non-minimum phase (NMP) system: definition; why Boost is NMP; step response initial wrong-direction transient; bandwidth limitation
- Buck vs. Boost Bode plots overlaid: candidate crossover frequencies; available phase margin

Conceptual sidebar (not a separate module – embedded naturally): state-space averaging concept; transfer function; pole-zero in view of energy; Bode plot reading; NMP system response.

✂ **Lab / Hands-On:** Buck simulation in LTspice – verify V_o/D from waveform; Python/Scilab Bode plot of $G_{vd}(s)$; Boost simulation – observe NMP step response; side-by-side Buck/Boost Bode overlay; phase margin measurement at candidate frequencies

Module 4 Assessment: Lab notebook: state-space derivation steps with physical interpretation; annotated Bode plots identifying LC pole, ESR zero, RHP zero; written answer – “Why does the Boost output dip before rising?”

COs: CO2, CO3 **Bloom Level:** BL3 (Ap), BL4 (An)

7.5 ⚡ Module 5: Gate Pulse Generation – From Analog to Professional IC (10 Hours)

Module Objective: Develop complete gate drive design capability progressing from analog principle to embedded implementation to professional IC deployment. Connect directly to Module 4's converter hardware.

🔑 Key Topics:

- Analog gate pulse: 555 timer astable/monostable; comparator-based PWM; ramp + comparator method
- Arduino PWM architecture: Timer register direct control; precision frequency and duty cycle programming
- Optical isolation: ground loop problem in converter gate drive; optocoupler principle; CTR; PC817 vs. 6N137; speed limitation
- TI UCC27524 dual 5A gate driver: datasheet reading – peak source/sink current, propagation delay, UVLO; gate resistor R_G design (switching loss vs. EMI trade-off); dead time; application in Boost chopper circuit
- ESP32 LEDC peripheral: precision PWM; Wi-Fi-enabled duty cycle monitoring; introduction to SiC/GaN gate drive requirements

✂ Lab / Hands-On: 555 timer PWM circuit; Arduino variable-frequency PWM verified on oscilloscope; PC817 optocoupler isolation implementation; UCC27524 application circuit in Boost gate drive; switching waveform comparison; ESP32 PWM with serial monitor

Module 5 Assessment: Lab notebook: annotated UCC27524 datasheet; R_G calculation; switching waveform photographs with physical explanation of rise time, delay, ringing

COs: CO3 **Bloom Level:** BL3 (Ap), BL6 (Cr)

Industry Connection: UCC27524 is deployed in real TI reference designs. Students who complete this module can directly use TI WEBENCH Power Designer and TI application notes – skills applicable on Day 1 of any power electronics engineering role.

8 Semester 6 – Part II (32 Contact Hours)

⚠ Entry Requirement for Semester 6: Students must have completed Semester 5 (Part-I) with satisfactory lab notebook submission and a minimum of 80% attendance. Semester 6 assumes fluency with: state-space modelling (Module 4), Bode plot reading, gate drive implementation (Module 5), and Arduino/ESP32 programming.

8.1 Module 6: Closed-Loop Control – Type-II and Type-III Compensator Design (10 Hours)

📌 Module Objective: Design professional-grade Type-II and Type-III compensators for Buck and Boost converters using the K-factor method. Implement in analog (op-amp) and digital (Arduino/ESP32) form. Verify loop gain and achieved stability margins. This module is the direct solution to the problem identified in Module 4.

Key Topics:

- Review of Module 4 Bode plots: problem statement – crossover frequency and phase margin deficit
- Type-I compensator: pure integrator; infinite DC gain; phase margin sacrifice; limitations
- Type-II compensator: structure (one zero, two poles with integrator); Bode shape; K-factor method – choose f_c , choose phase margin target, compute K , place zero and pole; op-amp component calculation
- Type-III compensator: structure (two zeros, three poles); handles Buck LC double pole and Boost ESR zero; K-factor extended; when to choose Type-II vs. Type-III
- Analog implementation: op-amp circuit realisation; layout considerations (noise, decoupling); integration with gate drive (Module 5) and converter (Module 4)
- Digital implementation: bilinear (Tustin) discretisation; implement on Arduino/ESP32; sampling rate selection; analog vs. digital transient response comparison
- Loop gain verification: injection method or simulation; achieved phase margin; closed-loop Bode; stability margin

✂ Lab / Hands-On: Type-II design for Buck: calculate components; Python Bode overlay (plant + compensator + loop); Type-III design for Boost; op-amp implementation and closed-loop Buck hardware test; discrete Type-II on Arduino; loop gain verification

📁 Module 6 Assessment: Lab notebook: K-factor design calculation; compensated Bode plot; measured step load response photograph with rise time and overshoot annotated; written comparison – Type-II vs. Type-III: when and why

COs: CO2, CO4 **Bloom Level:** BL4 (An), BL5 (Ev), BL6 (Cr)

⚡ Why Type-II/III, not PID? PID is a general-purpose controller. Type-II and Type-III are purpose-built for SMPS compensation – designed from the Bode plot, directly targeting crossover frequency and phase margin. Every professional SMPS design (Infineon, STMicroelectronics, TI, Renesas) uses this framework. TI WEBENCH Power Designer implements exactly this method.

8.2 ⚡ Module 7: Generalised Machine Theory and dq0 Transformation for Drive Control (12 Hours)

🚩 Module Objective: Answer the engineering question: “*We can now control a converter. Can we control a motor the same way? What stops us – and what mathematics do we need?*” Deliver the dq0 transformation as an engineering problem-solution narrative. Culminate in the FOC block diagram that visibly contains every module of this course.

🔑 Key Topics:

- DC motor as the naturally separated machine: field and armature orthogonality; $T = K\Phi I_a$; independent control
- Why AC machine control is hard: rotating flux; time-varying inductances in abc frame; coupled axes
- Generalised machine theory (minimum essentials – three equations only): flux-linkage voltage equation $V = Ri + d\lambda/dt$; energy in magnetic field; torque from co-energy; inductance matrix shown as reference table
- Reference frame theory: 2D rotation matrix; stationary, rotor, and synchronous reference frames; motivation for synchronous frame
- Park’s abc-to-dq0 transformation: matrix; d-axis = flux axis (equivalent field); q-axis = torque axis (equivalent armature); inverse transformation
- Validation on DC motor: apply dq0 to DC motor model; recover known equations – proof of physical correctness
- Application to BLDC: Hall sensor commutation as crude dq alignment; commutation torque ripple; continuous dq control
- Application to PMSM: dq voltage and torque equations; $i_d = 0$ strategy for maximum torque per ampere (MTPA); flux weakening concept
- FOC block diagram: speed controller; i_d/i_q current controllers (Type-II from Module 6 – explicitly identified); inverse Park; SVPWM/SPWM; encoder feedback
- **Course Convergence Session:** one block diagram containing all modules (see below)
- Industrial motor drive overview: TI InstaSPIN-FOC; Infineon iMOTION; path from this course to embedded drive development

✂ Lab / Hands-On: OpenModelica / Scilab PMSM dq model; step speed command; observe i_d , i_q , torque, speed responses; identify Type-II current controller blocks in the simulation; student whiteboard presentation of complete FOC diagram

Module 7 Assessment: Lab notebook: dq transformation of a DC motor showing recovery of known equations; annotated FOC block diagram identifying which module each block was developed in; oral explanation – “Why can we not just use a single PID on an AC motor stator voltage?”

COs: CO5 **Bloom Level:** BL2 (Un), BL3 (Ap)

★ Course Convergence – The Climax of the Two-Semester Journey:

Session 11 of Module 7 presents **one block diagram containing the entire course:**

- Gate pulse (Module 5) → Inverter circuit (Modules 3, 4)
- Type-II i_d and i_q current controllers (Module 6)
- Inverse Park transform → abc reference voltages
- dq0 forward transform ← current feedback (Module 7)
- Speed controller output → i_q^* reference
- Every inductor current waveform follows $(1 - e^{-t})$ (Module 2)
- Every Bode analysis used Volt-second balance (Module 1)

This single diagram is the reason the course is designed the way it is. Plan Session 11 as a dedicated integrative lecture with student-led whiteboard presentation.

8.3 ⚡ Module 8: Real-World Validation – OpenModelica and Hardware-in-the-Loop (10 Hours)

Module Objective: Bring the two-semester journey to a tangible, measurable conclusion by building and validating a complete motor drive system in OpenModelica simulation and/or HIL / PIL environment. The final demonstration requires students to explain every sub-block from Module 1 through Module 7.

🔑 Key Topics:

- OpenModelica environment and Modelica Standard Library; RC/RL circuit model – verify vs. Module 1 hand calculation
- Buck and Boost models using power electronics library: verify V_o/D curve vs. Module 4 state-space results
- Closed-loop converter: add Type-II/III compensator block (Module 6); step load and step reference; verify transient specs
- PMSM dq model using Modelica Machines library: FOC controller block; speed step, torque step, load disturbance
- HIL Option A: controller on Arduino/ESP32; plant in OpenModelica real-time; serial data exchange
- PIL Option B: controller code in simulation loop; real converter hardware as plant

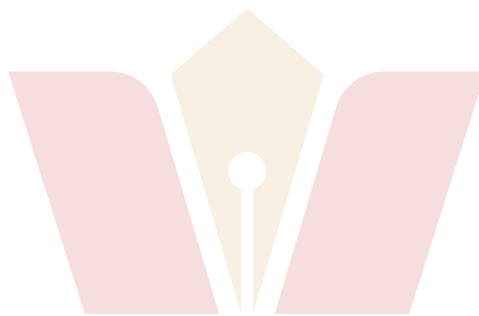
- Final assessed demonstration: each student presents complete drive system and explains every sub-block

✂ **Lab / Hands-On:** OpenModelica simulations building progressively from RC circuit (Module 1) to full FOC PMSM drive (Module 7); HIL or PIL setup; final oral presentation

📁 **Module 8 Assessment:** Final practical demonstration (assessed by panel): build or run a specified system; explain every design decision and observed waveform; connect back to exponential family and Volt-second balance as foundation

COs: CO6 **Bloom Level:** BL5 (Ev), BL6 (Cr)

✂ **Primary Tool – Why OpenModelica:** OpenModelica is free, open-source, runs on Linux (Manjaro compatible), uses the Modelica standard (deployed in automotive and aerospace industry), and includes motor, converter, and control library components. Students can continue using it independently after the course. BLDCSim supplements OpenModelica for BLDC-specific sessions.



9 Complete Module Summary

Table 6: Two-Semester Module Overview

Mod.	Title	Sem.	Hrs	COs	Bloom (A-K)	Role
1	Network Theorems and R, L, C Energy Properties	5	8	CO1, CO2	BL1, BL2, BL4	Foundation
2	The Exponential Family – One Curve, Many Worlds	5	8	CO1, CO2	BL1, BL2	Spine
3	Linear Voltage Regulator – Design and Fabrication	5	6	CO3	BL3, BL6	First Build
4	SMPS Buck/Boost – Open Loop Analysis	5	10	CO2, CO3	BL3, BL4	Problem ID
5	Gate Drive – Analog to UCC27524	5	10	CO3	BL3, BL6	Toolkit
Semester 5 Sub-total			42			
6	Type-II / Type-III Compensator Design	6	10	CO2, CO4	BL4, BL5, BL6	Solution
7	Generalised Machine Theory + dq0 + FOC	6	12	CO5	BL2, BL3	Generalisation
8	OpenModelica and HIL/PIL Validation	6	10	CO6	BL5, BL6	Climax
Semester 6 Sub-total			32			
Grand Total			74	CO1–CO6	BL1–BL6	Complete

10 CO – PO – PSO Mapping

❗ NBA Compliance Note: As per the current NBA criteria for B.Tech programmes, there are **11 Programme Outcomes (POs, PO1–PO11)** and **3 Programme Specific Outcomes (PSOs)**. Mapping strength: **3** = High, **2** = Moderate, **1** = Low, – = No mapping. Cells are colour-coded: **3** **2** **1**

PO Definitions (NBA, B.Tech – 11 POs):

PO1: Engineering Knowledge

PO2: Problem Analysis

PO3: Design / Development of Solutions

PO4: Conduct Investigations of Complex Problems

PO5: Modern Tool Usage

PO6: The Engineer and Society

PO7: Environment and Sustainability

PO8: Ethics

PO9: Individual and Team Work

PO10: Communication

PO11: Project Management and Finance

PSO Definitions (B.Tech EE, BCREC):

PSO1: Apply knowledge of electrical machines, power systems, and power electronics to analyse and design electrical energy systems

PSO2: Use simulation and embedded tools to implement and validate electrical engineering solutions

PSO3: Demonstrate awareness of energy efficiency, sustainability, and emerging technologies in the electrical engineering domain

Table 7: CO – PO – PSO Mapping Matrix (NBA: 11 POs + 3 PSOs)

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	1	2	2	–	–	1	–	2	–	3	1	1
CO2	3	3	2	3	3	–	–	1	–	1	–	3	3	1
CO3	2	2	3	2	3	1	–	2	2	2	1	3	3	1
CO4	3	3	3	3	3	–	–	1	1	2	2	3	3	1
CO5	3	3	2	2	3	1	–	1	–	2	–	3	3	2
CO6	3	3	3	3	3	1	1	2	2	2	2	3	3	2
Avg.	2.83	2.67	2.33	2.50	2.83	0.33	0.17	0.83	0.50	1.50	0.50	3.00	2.67	0.67

Strongly mapped POs: PO1 (Engineering Knowledge), PO2 (Problem Analysis), PO3 (Design of Solutions), PO5 (Modern Tool Usage). All three PSOs are addressed, with PSO1 and PSO2 mapped at high strength across all COs.

11 Assessment Structure

i Assessment Philosophy: This course carries no written examination. Understanding is demonstrated by doing. Assessment is continuous, practical, and oral – consistent with the hands-on nature of the course and the requirements for CO attainment evidence under NBA Criterion 3.

Table 8: Assessment Components and Weightage

Sl.	Component	Weight	Description
1	Lab Notebook / Module Report	30%	Submitted after each module; includes circuit diagrams, Python/Scilab plots, waveform photographs, annotated datasheets, and one physical interpretation paragraph per result. Evidence of CO attainment.
2	Practical Demonstration	20%	End of each semester: student builds a specified circuit, operates it, and explains every design choice and observed waveform before a faculty panel. (Mini Project)
3	Viva Voce	10%	10–15 minute oral examination per student; questions target physical understanding, not formula recall. Example: “Why does the Boost converter output dip before rising?” – answer must reference RHP zero, not just recite the term.
4	End semester open book examination	40%	End semester an open book examination (No Provision of re-examination)

Passing Criterion: Minimum 50% aggregate; minimum **80% attendance** in contact hours (college/gazette holidays are excluded; collective absence has no special status – see Section 5). Certificate of completion issued by the Department of Electrical Engineering, BCREC, on successful completion of both semesters.

12 NBA, NAAC, and AICTE Compliance

12.1 NBA Compliance

Table 9: NBA Criterion-wise Compliance

NBA Criterion	How This Course Addresses It
Criterion 3 – CO definition	Six measurable COs with Anderson-Krathwohl Bloom levels assigned; action verbs (Identify, Analyse, Apply, Design, Explain, Build) directly map to cognitive process levels
Criterion 3 – CO-PO-PSO mapping	Full 6×14 matrix (11 POs + 3 PSOs) with strength values
Criterion 3 – CO attainment	Lab notebooks, practical demonstrations, and viva voce serve as direct CO attainment measurement tools
Criterion 4 – Student performance	Module-wise continuous assessment with defined rubrics
Criterion 5 – Faculty contribution	Developed and delivered by Ph.D. faculty with active SCI/ESCI journal publications and ongoing research in power systems, SMPS, and BESS
Criterion 6 – Facilities	Delivered in Sim Lab-III, EE Dept., BCREC, with dedicated computer and hardware facilities; Tesla Research Lab available as extended venue
Criterion 9 – Learning environment	Small batch (20), lab-integrated delivery, industry tools (UCC27524, OpenModelica, Arduino, ESP32), real hardware implementation

12.2 NAAC Compliance

Table 10: NAAC Criterion-wise Compliance

NAAC Criterion	How This Course Addresses It
2.3 – Teaching-Learning	Experiential, collaborative, problem-based; lab-integrated; industry-standard tools; reconnection pedagogy
2.4 – Teacher Quality	Ph.D. faculty with active SCI/ESCI journal publications, funded research, and ongoing industry-relevant projects; live research informing teaching
2.6 – Student Performance	CO attainment documented through lab notebooks, demonstrations, viva; module-level continuous tracking
3.3 – Research Culture	Course directly rooted in faculty research on SMPS optimisation, BESS, and FOC; students exposed to open problems
3.4 – Extension Activities	Certificate course creates student project pipeline into Tesla Research Lab and industry-relevant final year projects
5.1 – Student Support	Mentorship-aligned; directly improves placement readiness in power electronics and EV sector

12.3 AICTE Compliance

Table 11: AICTE Guideline Compliance

AICTE Guideline	Compliance
Add-On course definition	Fully compliant: beyond regular curriculum, practical focus, certificated, industry-relevant
Contact hour counting	Integrated lecture cum hands-on; each session counts under AICTE L-T-P norm; no artificial separation
Batch size	20 students – within recommended range; promotes quality
OBE implementation	Six COs, Anderson-Krathwohl Bloom levels, CO-PO-PSO matrix, CO attainment measurement
Industry relevance	Type-II/III compensators, UCC27524, ESP32, Open-Modelica, HIL – direct industry application
NEP 2020 multidisciplinary	Connects mathematics (exponential family), physics (energy in fields), EE (machines, power electronics), and computing (Python, embedded systems)
Employability	Gate drive design, datasheet literacy, compensator design, FOC – Day 1 applicable skills in power electronics and EV sector

13 Impact Summary

Student-Level Impact

- ✔ Unified mathematical framework connecting every EE subject through the exponential family – compartmentalisation eliminated permanently
- ✔ Datasheet reading capability (LM317, UCC27524) – demanded on Day 1 of any power electronics role, absent from standard curriculum
- ✔ Type-II/III compensator design – directly applicable to SMPS design positions (Infineon, STMicroelectronics, TI, Renesas)
- ✔ FOC / motor drive fluency – directly applicable to EV drives, robotics, and servo systems; aligned with National EV Mission skill needs
- ✔ OpenModelica hands-on – the Modelica standard is used in automotive and aerospace industry
- ✔ Embedded + power hardware integration (Arduino, ESP32) – IoT-driven smart energy systems
- ✔ Lab notebook and viva assessment builds technical communication and documentation skills valued by industry and postgraduate admission committees

Departmental and Institutional Impact

- ✔ Strengthens NBA Criterion 3 CO attainment data with structured add-on course evidence
- ✔ Provides NAAC 2.3 and 3.4 documentation of innovative, research-linked pedagogy
- ✔ Sim Lab-III gains structured usage as a dedicated add-on course laboratory; Tesla Research Lab serves as an advanced extended venue for research-connected sessions
- ✔ Creates talent pipeline for final-year projects in power electronics, motor drives, and BESS optimisation
- ✔ Distinguishes BCREC in the MAKAUT ecosystem as an institution offering industry-depth add-on courses
- ✔ Replicable to ME and ECE departments with minor domain adaptation
- ✔ Graduates with FOC and power electronics build skills are attractive to EV-sector recruiters – directly improves placement statistics in a high-growth sector

14 ✂ Tools, Software, and References

14.1 🖨 Software and Hardware Tools

Table 12: Tools Used in This Course

Sl.	Tool	Purpose in Course	Availability
1	Python 3 (NumPy, SciPy, Matplotlib, python-control)	Curve plotting, Bode analysis, state-space computation, interactive dashboards	Free; open-source
2	Scilab / Xcos	Signal flow, block diagram simulation	Free; open-source
3	LTspice XVII	SMPS circuit simulation (Modules 4, 6)	Free (Analog Devices)
4	OpenModelica	System-level simulation and HIL (Module 8)	Free; open-source
5	BLDCSim	BLDC-specific motor simulation (Modules 7, 8)	Free
6	Arduino IDE	Arduino programming (Modules 5, 6)	Free; open-source
7	Arduino-ESP32 / ESP-IDF	ESP32 PWM and IoT integration (Module 5)	Free; open-source
8	TI WEBENCH Power Designer	Type-II/III compensator cross-verification (Module 6)	Free (TI online tool)
9	Oscilloscope (lab)	Waveform measurement across all modules	Institutional
10	Breadboard and components	Circuit fabrication (Modules 3, 5, 6)	Sim Lab-III / Tesla Research Lab

14.2 📖 Key References

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15 Course Instructor

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♥ **Best Wishes for Your Learning Journey!**

“Tell me and I forget. Teach me and I remember. Involve me and I learn.”
– Benjamin Franklin

Prepared by Kingsuk

End of: Add-On Course Syllabus: EE-AOC-581 & EE-AOC-681

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Date: July 2026, Version V6

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